Exercise 2 – SQL Formulations

SQL Aggregate Functions & Operators

① SELECT DISTINCT department
FROM Students ~~table~~;

department
IT
HR
IT
Finance
HR

② SELECT department
AVG (age) AS avg_age
FROM Students
GROUP BY department;

department	avg_age
IT	20.5
HR	22
Finance	23

3. SELECT department,
COUNT(*) AS student_count
FROM Students
GROUP BY department
HAVING COUNT(*) > 1;

department	student_count
IT	2
HR	2

4. SELECT *
FROM students
WHERE age BETWEEN 21 AND 23;

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

5. SELECT department
FROM students
WHERE age > 21 AND department = 'IT' OR 'HR';

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6. SELECT COUNT(*) AS total_credits
FROM courses
GROUP BY department
HAVING COUNT(*) > 5;

department	total_credits
IT	11

7. SELECT *
FROM courses
WHERE Credits NOT '4';

course_id	course name	department	credits
101	SQL basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course_name
FROM courses
ORDER BY credits DESC
LIMIT 3;

course id	course_name	credits
102	Python	4
103	Data science	4
101	SQL Basics	3

9. SELECT MIN (grade) AS min_grade,
MAX (grade) AS max_grade,
AVG (grade) AS avg_grade
FROM enrollments;

min_grade	max_grade	avg_grade
78	90	84.6

10. SELECT COUNT (student_id) AS course_id,
(enrollment_id) AS enrollment_count
FROM enrollments;

course_id	enrollment_count
101	1
102	2
103	3
104	4
105	5

11. SELECT (salary) AS total_salary,
(bonus) AS total_bonus
FROM salaries;

department	total_salary	totell_bonus
IT	122 000	10 500
HR	109 000	7500
Finance	70000	6000

12. SELECT (salary) AS avg-salary
FROM salaries
WHERE salary > 50 000

department	avg_salary
IT	61 000
Finance	70 000

13. SELECT (salary) AS total_compensation
FROM salaries
WHERE salary AND bonus > 60 000;

~~department~~	employee_id	name	department	salary	bonus	total_compensation
	1	Tom	IT	60 000	5000	65000
	3	Spike	Finance	70 000	6000	76 000
	4	Tyre	IT	62000	5500	67500

14. SELECT (budget) AS total_budget,
(budget) AS avg_budget
FROM projects
WHERE avg_budget > 70 000;

department	total_budget	~~total~~ avg_budget
IT	270 000	135000
Finance	80 000	80000

15. SELECT *
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000 NOT IN ('marketing');

project_id	project_name	department	budget
2	payroll system	Finance	80 000